

Causal Inference

Lecture 3

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based partially on Prof. Arvid Sjölander's slides

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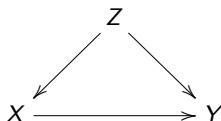
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- 3 Estimation without regression models
 - Standardization
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- In ideal randomized trials we have exchangeability:

$$(Y_0, Y_1) \perp\!\!\!\perp X$$

- As a consequence, association = causation



- In observational studies we usually don't have exchangeability, because of confounding
- We may achieve conditional exchangeability by controlling for an appropriate set of covariates:

$$(Y_0, Y_1) \perp\!\!\!\perp X \mid Z$$

- When Z is binary or categorical with few levels we can control for Z by stratification
 - conceptually simple
 - computationally simple
 - does not require any modeling assumptions
- Stratification gives one causal effect for each stratum - conditional (subpopulation) effects
 - e.g. stratification on sex gives the effect for men and women separately

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- Sometimes, it may be desirable to estimate the causal effect for the whole population - a **marginal** causal effect
 - easier to interpret and communicate one marginal effect than several conditional effects
 - randomized trials give marginal effects, and we may want to make results from observational studies comparable
 - we may want to consider future interventions to the whole population, rather than to subgroups
- Two estimation methods for marginal effects: standardization and inverse probability weighting (IPW)

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Example (recap)

	$Z = 0$		$Z = 1$	
	$Y = 0$	$Y = 1$	$Y = 0$	$Y = 1$
$X = 0$	270	30	120	80
$X = 1$	180	20	60	240

- Assume conditional exchangeability, given Z , and compute the conditional causal risk ratio, given Z , for $Z = 1$ and $Z = 0$.
- Where in the computation do you use the assumption of exchangeability?

	$Z = 0$		$Z = 1$	
	$Y = 0$	$Y = 1$	$Y = 0$	$Y = 1$
$X = 0$	270	30	120	80
$X = 1$	180	20	60	240

Conditional causal risk ratio, given $Z = 0$:

$$\begin{aligned}
 & \frac{p(Y_1 = 1|Z = 0)}{p(Y_0 = 1|Z = 0)} \stackrel{(\gamma_0, \gamma_1) \perp\!\!\!\perp X|Z}{=} \frac{p(Y_1 = 1|X = 1, Z = 0)}{p(Y_0 = 1|X = 0, Z = 0)} \\
 &= \frac{p(Y = 1|X = 1, Z = 0)}{p(Y = 1|X = 0, Z = 0)} \\
 &= \frac{20/200}{30/300} = 1
 \end{aligned}$$

	$Z = 0$		$Z = 1$	
	$Y = 0$	$Y = 1$	$Y = 0$	$Y = 1$
$X = 0$	270	30	120	80
$X = 1$	180	20	60	240

Conditional causal risk ratio, given $Z = 1$:

$$\begin{aligned}
 & \frac{p(Y_1 = 1|Z = 1)}{p(Y_0 = 1|Z = 1)} \stackrel{(Y_0, Y_1) \perp\!\!\!\perp X|Z}{=} \frac{p(Y_1 = 1|X = 1, Z = 1)}{p(Y_0 = 1|X = 0, Z = 1)} \\
 &= \frac{p(Y = 1|X = 1, Z = 1)}{p(Y = 1|X = 0, Z = 1)} \\
 &= \frac{240/300}{80/200} = 2
 \end{aligned}$$

- Given $Z = 0$, the conditional causal risk ratio is equal to 1 (no effect)
- Given $Z = 1$, the conditional causal risk ratio is equal to 2
- Effect modification by (interaction with) Z

- The conditional causal risk ratio, given Z , is defined as

$$\frac{p(Y_1 = 1|Z)}{p(Y_0 = 1|Z)}$$

- The marginal causal risk ratio is defined as

$$\frac{p(Y_1 = 1)}{p(Y_0 = 1)}$$

- How can we estimate the marginal causal risk ratio, while controlling for Z ?

- If we have conditional exchangeability, given Z , then

$$p(Y_0 = 1) = \sum_Z p(Y = 1|X = 0, Z)p(Z)$$

$$p(Y_1 = 1) = \sum_Z p(Y = 1|X = 1, Z)p(Z)$$

- Law of total probability:

$$p(Y_x = 1) = \sum_Z p(Y_x = 1|Z)p(Z)$$

- Conditional exchangeability, given Z :

$$\sum_Z p(Y_x = 1|Z)p(Z) = \sum_Z p(Y_x = 1|X = x, Z)p(Z)$$

- Definition of potential outcomes:

$$\sum_Z p(Y_x = 1|X = x, Z)p(Z) = \sum_Z p(Y = 1|X = x, Z)p(Z)$$

$$p(Y_x = 1) = \sum_Z p(Y = 1|X = x, Z)p(Z)$$

	Z = 0		Z = 1	
	Y = 0	Y = 1	Y = 0	Y = 1
X = 0	270	30	120	80
X = 1	180	20	60	240

- Assume conditional exchangeability, given Z , and compute the marginal causal risk ratio

	Z = 0		Z = 1	
	Y = 0	Y = 1	Y = 0	Y = 1
X = 0	270	30	120	80
X = 1	180	20	60	240

$$\begin{aligned}
 p(Y_0 = 1) &\stackrel{(Y_0, Y_1) \perp\!\!\!\perp X|Z}{=} \sum_Z p(Y = 1|X = 0, Z)p(Z) \\
 &= \underbrace{30/300}_{p(Y=1|X=0, Z=0)} \times \underbrace{500/1000}_{p(Z=0)} + \underbrace{80/200}_{p(Y=1|X=0, Z=1)} \times \underbrace{500/1000}_{p(Z=1)} = 0.25
 \end{aligned}$$

$$\begin{aligned}
 p(Y_1 = 1) &\stackrel{(Y_0, Y_1) \perp\!\!\!\perp X|Z}{=} \sum_Z p(Y = 1|X = 1, Z)p(Z) \\
 &= \underbrace{20/200}_{p(Y=1|X=1, Z=0)} \times \underbrace{500/1000}_{p(Z=0)} + \underbrace{240/300}_{p(Y=1|X=1, Z=1)} \times \underbrace{500/1000}_{p(Z=1)} = 0.45
 \end{aligned}$$

$$\frac{p(Y_1 = 1)}{p(Y_0 = 1)} = \frac{0.45}{0.25} = 1.8$$

- In our example we observed that

$$p(Z = 0) = p(Z = 1) = 0.5$$

and

$$\frac{p(Y_1 = 1|Z = 0)}{p(Y_0 = 1|Z = 0)} = 1$$

$$\frac{p(Y_1 = 1|Z = 1)}{p(Y_0 = 1|Z = 1)} = 2$$

$$\frac{p(Y_1 = 1)}{p(Y_0 = 1)} = 1.8$$

- The marginal effect is generally not equal to the average of the conditional effects, even if all levels of Z are equally probable .
- This property is known as non-collapsibility: Non-collapsibility occurs when a marginal measure of association (e.g., an odds ratio, risk ratio) is not equal to a weighted average of the conditional measures of association across levels of a covariate(s).
- Exception: the causal risk difference

- The marginal effect is generally not equal to the average of the conditional effects, even if these are constant across levels of Z
 - e.g. the conditional causal odds ratio may be equal to 3 for both males ($Z = 1$) and females ($Z = 0$),
 - but the marginal causal odds ratio may be equal to 1.5
- This is often referred to as 'non-collapsibility'
- Exceptions: the causal risk difference and the causal risk ratio (collapsible)

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- Like standardization, inverse probability weighting (IPW) is a method to estimate marginal causal effects
- Without modeling assumptions, IPW gives the same result as standardization
- IPW may give different results, and may sometimes be preferable, when using regression models
 - more later

- **Step 1:** compute the probability $p(X|Z)$ for each level of the exposure X and confounder Z
- **Step 2:** assign a weight to each subject, equal to

$$W = \frac{1}{p(X|Z)}$$

where $p(X|Z)$ is the probability of the subject's observed level of X , given the subject's observed level of Z

- for instance, suppose that $p(X = 1|Z = 1) = 0.2$
 - each subject with $(X = 1, Z = 1)$ is then counted as $1/0.2 = 5$ subjects
- **Step 3:** use $p(Y = 1|X = 0)$ and $p(Y = 1|X = 1)$ in the weighted sample as estimates of $p(Y_0 = 1)$ and $p(Y_1 = 1)$

Example

	$Z = 0$		$Z = 1$	
	$Y = 0$	$Y = 1$	$Y = 0$	$Y = 1$
$X = 0$	270	30	120	80
$X = 1$	180	20	60	240

- **Step 1:** compute the probability $p(X|Z)$ for each level of the exposure X and confounder Z

	$Z = 0$		$Z = 1$	
	$Y = 0$	$Y = 1$	$Y = 0$	$Y = 1$
$X = 0$	270	30	120	80
$X = 1$	180	20	60	240

$$p(X = 0|Z = 0) = 300/500 = 0.6$$

$$p(X = 1|Z = 0) = 200/500 = 0.4$$

$$p(X = 0|Z = 1) = 200/500 = 0.4$$

$$p(X = 1|Z = 1) = 300/500 = 0.6$$

	$Z = 0$		$Z = 1$	
	$Y = 0$	$Y = 1$	$Y = 0$	$Y = 1$
$X = 0$	270	30	120	80
$X = 1$	180	20	60	240

- **Step 2:** assign a weight to each subject i , equal to

$$W = \frac{1}{p(X|Z)}$$

	$Z = 0$		$Z = 1$	
	$Y = 0$	$Y = 1$	$Y = 0$	$Y = 1$
$X = 0$	$\frac{270}{0.6} = 450$	$\frac{30}{0.6} = 50$	$\frac{120}{0.4} = 300$	$\frac{80}{0.4} = 200$
$X = 1$	$\frac{180}{0.4} = 450$	$\frac{20}{0.4} = 50$	$\frac{60}{0.6} = 100$	$\frac{240}{0.6} = 400$

	$Z = 0$		$Z = 1$	
	$Y = 0$	$Y = 1$	$Y = 0$	$Y = 1$
$X = 0$	450	50	300	200
$X = 1$	450	50	100	400

- **Step 3:** use $p(Y = 1|X = 0)$ and $p(Y = 1|X = 1)$ in the weighted sample as estimates of $p(Y_0 = 1)$ and $p(Y_1 = 1)$

	Z = 0		Z = 1	
	Y = 0	Y = 1	Y = 0	Y = 1
X = 0	450	50	300	200
X = 1	450	50	100	400

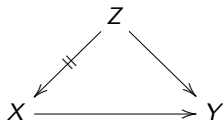
$$p(Y_0 = 1) = p(Y = 1|X = 0) = \frac{50 + 200}{450 + 50 + 300 + 200} = 0.25$$

$$p(Y_1 = 1) = p(Y = 1|X = 1) = \frac{50 + 400}{450 + 50 + 100 + 400} = 0.45$$

$$\frac{p(Y_1 = 1)}{p(Y_0 = 1)} = \frac{0.45}{0.25} = 1.8$$

- Same as with standardization

- IPW breaks the association between X and Z



- As a consequence, Z is not a confounder in the weighted sample
- If Z is the only confounder in the original sample, then there is no confounder at all in the weighted sample
- Thus, in the weighted sample we have exchangeability

Example

- Original sample:

	$Z = 0$		$Z = 1$	
	$Y = 0$	$Y = 1$	$Y = 0$	$Y = 1$
$X = 0$	270	30	120	80
$X = 1$	180	20	60	240

- Weighted sample:

	$Z = 0$		$Z = 1$	
	$Y = 0$	$Y = 1$	$Y = 0$	$Y = 1$
$X = 0$	450	50	300	200
$X = 1$	450	50	100	400

- Verify that X and Z are associated in the original sample, but independent in the weighted sample*

- Original sample:

	$Z = 0$		$Z = 1$	
	$Y = 0$	$Y = 1$	$Y = 0$	$Y = 1$
$X = 0$	270	30	120	80
$X = 1$	180	20	60	240

$$p(X = 1|Z = 0) = 200/500 = 0.4$$

$$p(X = 1|Z = 1) = 300/500 = 0.6$$

- $p(X = 1|Z = 0) \neq p(X = 1|Z = 1)$ so X and Z are associated

- Weighted sample:

	$Z = 0$		$Z = 1$	
	$Y = 0$	$Y = 1$	$Y = 0$	$Y = 1$
$X = 0$	450	50	300	200
$X = 1$	450	50	100	400

$$p(X = 1|Z = 0) = \frac{500}{1000} = 0.5$$

$$p(X = 1|Z = 1) = \frac{500}{1000} = 0.5$$

- $p(X = 1|Z = 0) = p(X = 1|Z = 1)$ so X and Z are independent

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- Suppose we carry out an observational study to estimate the causal effect of AZT on infection risk for AIDS patients
- 10,000 subjects enrolled
- Baseline measures:
 - CD4 count (Z ; counts/ μl)
 - AZT level (X ; '0' for 'untreated', '1' for 'treated')
- At end of follow up we measure:
 - infection status (Y ; '0' for no infection, '1' for infection)
- No deaths or drop-outs during follow-up

```
aids = pd.read_csv("aids.txt", sep=" ", header=0)
print(aids.head(10))
```

	Z	X	Y
0	305	0	1
1	359	1	0
2	429	0	0
3	293	1	0
4	346	1	0
5	357	1	0
6	385	0	0
7	338	0	0
8	449	0	0
9	343	0	0

```
# Create contingency table
table = pd.crosstab(aids['X'], 1 - aids['Y'])

# Extract counts and totals
count = table.iloc[:, 0].values # "Successes" for both groups
nobs = table.sum(axis=1).values # Total observations for both groups

# Proportions for verification
print("Proportions:", count / nobs)
```

Proportions: [0.57661788 0.07854008]

- Infections much more common among untreated than among treated; statistically significant association

- Subjects with low CD4 count are more likely to get AZT, and more likely to get infections
- Arguable, CD4 count is an important confounder that we need to control for
- But in the data, very few subjects have the same CD4 count
 - stratification on CD4 count is not feasible
- Let's use a regression model

The logistic regression model

- Since the outcome is binary, it is natural to use the logistic regression model

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Why do we need GLM?

- Until now, we assumed that the error term was distributed approximately normally. Hence, the conditional distribution of the response variable Y was also approximately normal.
- In many situations, this assumption is absolutely inadequate, e.g., where the response variable is dichotomous (failure/success, $Y \in \{0, 1\}$).
- In GLMs' framework, the linearity is preserved through a link function g , hence the name “generalized” linear model.
- GLMs have a closed analytical theory of estimation and inference.

Construction

- The distribution of the response variable Y , given the explanatory variables X , belongs to the exponential family (e.g., Bernoulli, Exponential, Chi-squared, Poisson and more).
- The conditional expectation of Y given $X = x$ is linear via g ,

$$\mathbb{E}[Y|X] = g(\beta_0 + \beta_1 x_1 + \cdots \beta_p x_p),$$

hence

$$g^{-1}(\mathbb{E}[Y|X]) = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p,$$

where g is the link function.

- To use a GLM, we need to specify the distribution and the link function.
- Any invertible and smooth function g can be a link function. However, the default option is the so-called “canonical link function” which is, in a sense, the most “natural” link function.

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Construction

- We have a binary response variable Y , that can be coded as 0 and 1.
- Assume that we have a data set of n (conditionally) independent realizations, $(y_i, x_{1i}, \dots, x_{pi})$ of some unknown data generating process (distribution) $g(x; \beta)$.
- $g(x_i; \beta)$ is the conditional probability of success, $g(x_i; \beta) \equiv p_i(\beta)$. Its explicit form is given by

$$\mathbb{E}[Y|X_i = x_i] = p_i(\beta) = (1 + \exp\{-(\beta_0 + \beta_1 x_{1i} + \dots + \beta_p x_{pi})\})^{-1}.$$

- The linearity is w.r.t. the $\ln$ of the Odds $_i = p_i(\beta)/(1 - p_i(\beta))$, i.e.,

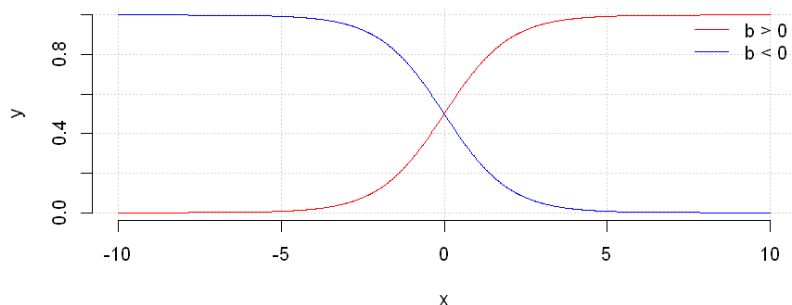
$$g^{-1}(\mathbb{E}[Y|X_i = x_i]) = \ln\left(\frac{p_i(\beta)}{1 - p_i(\beta)}\right) = \beta_0 + \beta_1 x_{1i} + \dots + \beta_p x_{pi},$$

the left-hand-side is called the logit function of $p_i(\beta)$, hence the name “logistic” regression.

Construction

- The Image (range) of $p_i(\beta) \equiv g(x_i; \beta)$ is $[0, 1]$ as expected from a probability measure.

$$p(y|x) = (1 + e^{-bx})^{-1}$$



Odds ratio

- Assume for illustration purposes that we have only one explanatory variable X , such that $\ln(\text{Odds}(x_i)) = \beta_0 + \beta_1 x_i$.
- In simple linear regression, β_1 is the expected change in the response variable Y for a change of one unit in X .
- In logistic regression, β_1 is the $\ln$ of the odds ratio

$$\beta_1 = \ln(\text{Odds}(x_i + 1)) - \ln(\text{Odds}(x_i)) = \ln \left(\frac{\text{Odds}(x_i + 1)}{\text{Odds}(x_i)} \right).$$

- Therefore e^{β_1} is the odds ratio (OR) of change in one unit in x .
- For example, if x designates age in years, then $e^{\beta_1 \Delta x}$ is the OR for a change of Δx years.

Maximum likelihood

- Unlike in linear regression, Y is not normal and not continuous. Hence, the OLS method is irrelevant.
- The estimation in GLMs is done by maximizing the likelihood function.
- Intuitively, maximizing the likelihood function is obtaining the values of the parameters that maximize the (joint) probability function of the given data under a certain parametric model.
- Formally, each observation i is a realization of Bernoulli random variable Y_i whose probability function is $p_i^{Y_i}(1 - p_i)^{1-Y_i}$, thus for n independent realizations the joint probability is given by

$$L(p) = \prod_{i=1}^n p_i^{y_i}(1 - p_i)^{1-y_i},$$

which is the likelihood function of $(y_1, \dots, y_n)$.

Maximum likelihood

- Each p_i depends on a set of explanatory variables $\mathbf{x}_i$ via inverse logit function, i.e., $p_i(\beta) = (1 + \exp\{-\beta^T \mathbf{x}_i\})^{-1}$, hence the likelihood function w.r.t. the coefficients β is

$$L(\beta) = \prod_{i=1}^n \frac{1}{(1 + e^{-\beta^T \mathbf{x}_i})^{y_i}} \left(\frac{e^{-\beta^T \mathbf{x}_i}}{1 + e^{-\beta^T \mathbf{x}_i}} \right)^{1-y_i}.$$

- Maximizing $L(\beta)$ is done by solving a system of $p + 1$ nonlinear equations. This is done numerically by iterative methods like the Newton-Raphson.
- The maximum likelihood estimators (MLEs) in regular settings are asymptotically normal and unbiased. Namely, for large sample sizes, the inference on the coefficients estimators $\hat{\beta}$ is done in the same manner as in normal linear regression.

$$\text{logit}\{p(Y = 1|X, Z)\} = \alpha + \beta X + \gamma Z$$

What are the interpretations of α , β , and γ ?

$$\begin{aligned}\alpha &= \text{logit}\{p(Y = 1|X = 0, Z = 0)\} \\ &= \log \left\{ \frac{p(Y = 1|X = 0, Z = 0)}{p(Y = 0|X = 0, Z = 0)} \right\}\end{aligned}$$

$$\begin{aligned}\beta &= \text{logit}\{p(Y = 1|X = 1, Z)\} - \text{logit}\{p(Y = 1|X = 0, Z)\} \\ &= \log \left\{ \frac{p(Y = 1|X = 1, Z)}{p(Y = 0|X = 1, Z)} / \frac{p(Y = 1|X = 0, Z)}{p(Y = 0|X = 0, Z)} \right\}\end{aligned}$$

$$\begin{aligned}\gamma &= \text{logit}\{p(Y = 1|X, Z + 1)\} - \text{logit}\{p(Y = 1|X, Z)\} \\ &= \log \left\{ \frac{p(Y = 1|X, Z + 1)}{p(Y = 0|X, Z + 1)} / \frac{p(Y = 1|X, Z)}{p(Y = 0|X, Z)} \right\}\end{aligned}$$

$$\text{logit}\{p(Y = 1|X, Z)\} = \alpha + \beta X + \gamma Z$$

- If we have conditional exchangeability, given Z , then β is the conditional causal log odds ratio, given Z

$$\beta = \log \left\{ \frac{p(Y_1 = 1|Z)}{p(Y_1 = 0|Z)} / \frac{p(Y_0 = 1|Z)}{p(Y_0 = 0|Z)} \right\}$$

$$\text{logit}\{p(Y = 1|X, Z)\} = \alpha + \beta X + \gamma Z$$

- *What assumptions does this model make?*

$$\text{logit}\{p(Y = 1|X, Z)\} = \alpha + \beta X + \gamma Z$$

- No effect modification by (interaction with) Z
 - the log odds ratio comparing treated and untreated, is assumed equal across levels of CD4 count
- Linear 'effect' of Z
 - the log odds ratio comparing two groups who differ with 1 unit in CD4 count, is the same regardless of the absolute CD4 count levels in the groups

- **All models are wrong**
 - but if the model is approximately correct, then our conclusions are approximately valid
- Assumptions that we make should ideally be justified by both
 - subjects matter knowledge, and
 - diagnostic tests with data

Fitting the model

```
import statsmodels.api as sm
```

```
# Define the dependent (Y) and independent variables (X, Z)
X = aids[["X", "Z"]] # Independent variables
X = sm.add_constant(X) # Add intercept term
y = aids["Y"] # Dependent variable
```

```
# Fit the logistic regression model
model = sm.Logit(y, X)
result = model.fit()
```

```
=====
Dep. Variable:          Y      No. Observations:          10000
Model:                Logit    Df Residuals:              9997
Method:                MLE     Df Model:                  2
Date:                  Wed, 27 Nov 2024    Pseudo R-squ.:        0.2576
Time:                  02:44:41    Log-Likelihood:       -4855.4
converged:              True     LL-Null:              -6539.9
Covariance Type:        nonrobust    LLR p-value:          0.000
=====
```

coef	std err	z	P> z	[0.025	0.975]
const	4.2334	0.206	20.506	0.000	3.829 4.638
X	-3.2531	0.071	-46.056	0.000	-3.392 -3.115
Z	-0.0107	0.001	-19.265	0.000	-0.012 -0.010

```
=====
```

• Interpretation?

```
> fit <- glm(formula=Y~X+Z+X*Z, family=binomial,  
data=aids)  
> summary(fit)
```

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	3.9586319	0.2282073	17.347	< 2e-16 ***
X	-2.0389789	0.4537132	-4.494	6.99e-06 ***
Z	-0.0099542	0.0006154	-16.176	< 2e-16 ***
X:Z	-0.0037779	0.0014037	-2.691	0.00712 **

- *Interpretation?*

What to report?

```
# Define the dependent (Y) and independent variables (X, Z, X*Z)
aids["X:Z"] = aids["X"] * aids["Z"] # Create interaction term
X = aids[["X", "Z", "X:Z"]] # Independent variables
X = sm.add_constant(X) # Add intercept term
y = aids["Y"] # Dependent variable
# Fit the logistic regression model
model = sm.Logit(y, X)
result = model.fit()
# Display the summary of the model
print(result.summary())
```

```
=====
Dep. Variable:                Y      No. Observations:          10000
Model:                        Logit    Df Residuals:             9996
Method:                       MLE      Df Model:                 3
Date:                        Wed, 27 Nov 2024    Pseudo R-squ.:          0.2581
Time:                        02:50:43    Log-Likelihood:         -4851.7
converged:                     True      LL-Null:                 -6539.9
Covariance Type:              nonrobust    LLR p-value:             0.000
=====
```

coef	std err	z	P> z	[0.025	0.975]
const	3.9586	0.228	17.347	0.000	3.511 4.406
X	-2.0390	0.454	-4.494	0.000	-2.928 -1.150
Z	-0.0100	0.001	-16.176	0.000	-0.011 -0.009
X:Z	-0.0038	0.001	-2.691	0.007	-0.007 -0.001

```
=====
```


- The main effect together with the interaction term?
 - unintuitive for non-statisticians
 - cumbersome if many confounders and interaction terms
- The effect at the mean/median level of Z ?
 - possibly not very informative about the 'general' treatment effect, unless most subjects are close to the mean/median

$$p(Y_0 = 1) \text{ vs } p(Y_1 = 1)$$

- Arguably more intuitive than main effect + interaction term
- Can always be presented as one single number (e.g. one log odds ratio) regardless of the number of interactions
- Informative about the 'general' treatment effect, since it applies to the whole population

- If we have conditional exchangeability, given Z , then

$$p(Y_0 = 1) = \sum_Z p(Y = 1|X = 0, Z)p(Z)$$

$$p(Y_1 = 1) = \sum_Z p(Y = 1|X = 1, Z)p(Z)$$

- If Z is binary, then we can estimate $p(Y = 1|X = 0, Z)$ and $p(Y = 1|X = 1, Z)$ without modeling assumptions
- If Z is continuous, this non-parametric approach is not feasible
 - very few subjects for each observed level of Z
- But with a regression model we can 'extrapolate', to estimate both $p(Y = 1|X = 0, Z)$ and $p(Y = 1|X = 1, Z)$

Four steps for standardization with regression model

- **Step 1:** fit a regression model for the outcome
- **Step 2:** replace the factual level of X with 0 for each subject
- **Step 3:** use the fitted model to estimate $p(Y = 1|X, Z)$ for each subject, i.e. for each level of (X, Z)
- **Step 4:** average these estimates to obtain an estimate of $p(Y_0 = 1)$
- To estimate $p(Y_1 = 1)$, replace X with 1 in step 2

Code for standardization with regression model

```
# Step 1: Fit the logistic regression model
model = sm.Logit(y, X)
fit = model.fit()

# Step 2: Create a copy of the dataset for X=0
aids0 = aids.copy()
aids0["X"] = 0 # Set X to 0
aids0["X:Z"] = aids0["X"] * aids0["Z"] # Update interaction term

# Step 3: Predict probabilities for X=0
X0 = aids0[["X", "Z", "X:Z"]]
X0 = sm.add_constant(X0) # Add intercept
pred0 = fit.predict(X0)

# Step 4: Calculate the mean predicted probability for X=0
p0 = pred0.mean()
print("Mean predicted probability for X=0:", p0)
Mean predicted probability for X=0: 0.6090828480763144
```

Code for standardization with regression model, continued

```
# Step 1: Fit the logistic regression model
model = sm.Logit(y, X)
fit = model.fit()

# Step 2: Create a copy of the dataset for X=1
aids1 = aids.copy()
aids1["X"] = 1 # Set X to 1
aids1["const"] = 1

# Step 3: Predict probabilities for X=1
X1 = aids1[["const", "X", "Z"]]
X1["X:Z"] = aids1["X"] * aids1["Z"]

pred1 = fit.predict(X1)

# Step 4: Calculate the mean predicted probability for X=1
p1 = pred1.mean()
print("Mean predicted probability for X=1:", p1)
Mean predicted probability for X=1: 0.06333355735321514
```

The marginal causal log odds ratio

$$\begin{aligned}\hat{p}(Y_0 = 1) &= 0.6090828 \\ \hat{p}(Y_1 = 1) &= 0.06333356\end{aligned}$$

- We can use $\hat{p}(Y_0 = 1)$ and $\hat{p}(Y_1 = 1)$ to estimate of the marginal causal log odds ratio

$$\log \left\{ \frac{\hat{p}(Y_1 = 1)}{1 - \hat{p}(Y_1 = 1)} / \frac{\hat{p}(Y_0 = 1)}{1 - \hat{p}(Y_0 = 1)} \right\} = -3.14$$

$$\begin{aligned}\hat{p}(Y_0 = 1) &= 0.6090828 \\ \hat{p}(Y_1 = 1) &= 0.06333356\end{aligned}$$

- Once we have estimated $p(Y_1 = 1)$ and $p(Y_0 = 1)$ we can estimate any measure of marginal causal effect, e.g.

$$\text{causal risk difference} = \hat{p}(Y_1 = 1) - \hat{p}(Y_0 = 1) = -0.55$$

$$\text{causal risk ratio} = \hat{p}(Y_1 = 1) / \hat{p}(Y_0 = 1) = 0.10$$

even though the estimates were derived from a logistic regression model


```
> library(stdReg)
> fit <- glm(formula=Y~X+Z+X*Z, family=binomial,
data=aids)
> std.fit <- stdGlm(fit=fit, data=aids, X="X")
> summary(std.fit)
```

Formula: $Y \sim X + Z + X * Z$

Family: binomial

Link function: logit

Exposure: X

	Estimate	Std. Error	lower 0.95	upper 0.95
0	0.6091	0.00655	0.5962	0.6219
1	0.0633	0.00348	0.0565	0.0702

```

# For X=1
aids1 = aids.copy()
aids1["const"] = 1
aids1["X"] = 1
aids1["X:Z"] = aids1["X"] * aids1["Z"] # Update interaction term

X1 = aids1[["const", "X", "Z", "X:Z"]]
pred1 = fit.predict(X1) # Predict probabilities for X=1
mean_pred1 = pred1.mean() # Average probability for X=1

# Step 4: Display results
print(f"Standardized probabilities:")
print(f"X=0: {mean_pred0:.4f}")
print(f"X=1: {mean_pred1:.4f}")

```

```

Standardized probabilities:
X=0: 0.6091
X=1: 0.0633

```

```
> summary(object=std.fit, transform="logit",  
contrast="difference", reference=0)
```

Formula: $Y \sim X + Z + X * Z$

Family: binomial

Link function: logit

Exposure: X

Transform: logit

Reference level: $X = 0$

Contrast: difference

	Estimate	Std. Error	lower 0.95	upper 0.95
0	0.00	0.0000	0.00	0.00
1	-3.14	0.0644	-3.26	-3.01

```
# Load the dataset
aids = pd.read_csv("aids.txt", sep=" ")

# Compute interaction term manually
aids["X:Z"] = aids["X"] * aids["Z"]

# Create the design matrix (manual intercept)
X = aids[["X", "Z", "X:Z"]]
X.insert(0, "intercept", 1)
y = aids["Y"]

# Fit the logistic regression model
model = sm.Logit(y, X)
fit = model.fit()

# Predict log-odds for X=0
aids0 = aids.copy()
aids0["X"] = 0
aids0["X:Z"] = aids0["X"] * aids0["Z"]
X0 = aids0[["X", "Z", "X:Z"]]
X0.insert(0, "intercept", 1)
log_odds0 = fit.predict(X0, linear=True)

# Predict log-odds for X=1
aids1 = aids.copy()
aids1["X"] = 1
aids1["X:Z"] = aids1["X"] * aids1["Z"]
X1 = aids1[["X", "Z", "X:Z"]]
X1.insert(0, "intercept", 1)
log_odds1 = fit.predict(X1, linear=True)

# Calculate the average difference
difference = log_odds1.mean() - log_odds0.mean()
print(f"Difference in log-odds (X=1 vs X=0): {difference:.4f}")

Difference in log-odds (X=1 vs X=0): -3.3633
```

- 1 Recap - Exchangeability
- 2 Marginal effects
- 3 Estimation without regression models
 - Standardization
 - Inverse probability weighting
- 4 Estimation with regression models
 - Standardization
- 5 GLM
 - Introduction
 - Logistic regression
 - Inverse probability weighting

Three steps for IPW with regression model

- **Step 1:** fit a regression model for the exposure
- **Step 2:** use the fitted model to estimate the subject-specific weight

$$W = \frac{1}{p(X|Z)}$$

- **Step 3:** use $p(Y = 1|X = 0)$ and $p(Y = 1|X = 1)$ in the weighted sample as estimates of $p(Y_0 = 1)$ and $p(Y_1 = 1)$

Code for IPW with regression model

```
> #step 1
> fit <- glm(formula=X~Z, family=binomial,
data=aids)
> #step 2
> pred <- predict(object=fit, type="respons")
> w <- 1/(aids$X*pred+(1-aids$X)*(1-pred))
> #step 3 for x=0
> p0 <- weighted.mean(aids$Y[aids$X==0],
w[aids$X==0])
> p0
[1] 0.6086676
> #step 3 for x=1
> p1 <- weighted.mean(aids$Y[aids$X==1],
w[aids$X==1])
> p1
[1] 0.06261394
```

Code for IPW with regression model in Python

```
# Step 1: Load the saved DataFrame
aids_df = pd.read_csv("aids.txt", sep=" ")

# Step 2: Fit logistic regression
X = add_constant(aids_df['Z']) # Add intercept
y = aids_df['X']
logit_model = Logit(y, X).fit()

# Predict probabilities
aids_df['pred'] = logit_model.predict(X)

# Calculate weights
aids_df['w'] = 1 / (aids_df['X'] * aids_df['pred'] +
                   (1 - aids_df['X']) * (1 - aids_df['pred']))

# Step 3: Calculate weighted mean for X = 0
p0 = np.average(aids_df.loc[aids_df['X'] == 0, 'Y'],
               weights=aids_df.loc[aids_df['X'] == 0, 'w'])

# Step 3: Calculate weighted mean for X = 1
p1 = np.average(aids_df.loc[aids_df['X'] == 1, 'Y'],
               weights=aids_df.loc[aids_df['X'] == 1, 'w'])

p0, p1
(0.6086675565580764, 0.06261393809533125)
```


The marginal causal log odds ratio

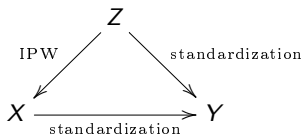
$$\begin{aligned}\hat{p}(Y_0 = 1) &= 0.6086676 \\ \hat{p}(Y_1 = 1) &= 0.06261394\end{aligned}$$

- We can use $\hat{p}(Y_0 = 1)$ and $\hat{p}(Y_1 = 1)$ to estimate of the marginal causal log odds ratio

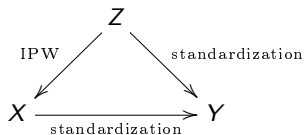
$$\log \left\{ \frac{\hat{p}(Y_1 = 1)}{1 - \hat{p}(Y_1 = 1)} / \frac{\hat{p}(Y_0 = 1)}{1 - \hat{p}(Y_0 = 1)} \right\} = -3.15$$

... but we can also use $\hat{p}(Y_0 = 1)$ and $\hat{p}(Y_1 = 1)$ to estimate any other measure of marginal causal effect

- e.g. the causal risk difference of risk ratio



- Standardization and IPW require different models
 - standardization requires an outcome model, describing how the outcome depends on the exposure and confounders
 - IPW requires an exposure model, describing how the exposure depends on the confounders
- If we know more about how the confounders affect the exposure, then more natural to use an exposure model
- If we know more about how the confounders affect the outcome, then more natural to use an outcome model



- Standardization gives an unbiased estimate if the outcome model is correct
- IPW gives an unbiased estimate if the exposure model is correct
- But if the model that we use is incorrect, then the obtained estimate is generally biased
 - model misspecification bias - not confounding bias
- It is possible to combine both models in **doubly robust** estimation
 - unbiased estimate if either model is correct, not necessarily both
 - two chances of unbiased estimate instead of only one

- Standard methods for confounding control (e.g. stratification and outcome regression models) give conditional (subpopulation) causal effects
 - provided that we have conditional exchangeability, given the measured confounders
- Sometimes, the marginal (population) causal effect may be a more relevant target parameter
- Marginal causal effects can be estimated with standardization and IPW
 - either non-parametrically, or with regression models